

PHENYLKETONURIA

AT A GLANCE

- **Autosomal recessive** disorder caused by **phenylalanine hydroxylase deficiency** (OMIM #261600).
- Gene located on chromosome **12q22-24.1**.
- Incidence: approximately **1 in 10,000 live births** among Europeans.
- **Clinical features:** mental retardation, generalized hypopigmentation, blond hair, blue eyes, eczematous rash, infantile vomiting, hyperactivity, seizures, mousy odor (due to phenylacetic acid).
- **Diagnosis:** confirmed by elevated serum phenylalanine levels; urinary ferric chloride test may be positive.
- **Treatment:** lifelong restriction of dietary phenylalanine.
- **Differential diagnosis:** albinism, Chediak-Higashi syndrome, scleroderma.
- **Prenatal diagnosis:** possible via chorionic villus sampling or amniocentesis, metabolite measurement, and DNA analysis.
- Biochemical phenotyping is now feasible in most newborns with hyperphenylalaninemia.

Epidemiology

The prevalence of phenylketonuria (PKU) varies significantly by region: - Highest incidence: **Ireland and Scotland** (1 in 4,000). - Lowest incidence: **Finland** (1 in 40,000). - Northern European average: **~1 in 10,000**. - United States: **~1 in 11,000** in the white population. - Among Black populations: **~1 in 50,000**.

Clinical Features

Classic manifestations of PKU typically emerge between **4 and 24 months of age**, despite normal appearance at birth.

Neurological Features:

- **Mental retardation** (marked), with irritability, hyperactivity, self-destructive behavior.
- **Seizures**, hypertonicity, exaggerated deep tendon reflexes.
- **Peculiar gait**, microcephaly.
- **Brain calcifications** and **cataracts** may occur.

Cutaneous Features:

- **Eczematous dermatitis** (20–50% of infants in first year), often indistinguishable from atopic dermatitis; may improve dramatically with dietary intervention.
- **Hypopigmentation**: pale skin, **blond hair**, **blue eyes** (in ~90% of patients), due to inhibition of tyrosinase by excess phenylalanine and its metabolites (e.g., phenylpyruvic acid, phenylacetic acid). - Pigmentation may partially reverse with dietary control. - Particularly striking in individuals of darker ethnic backgrounds (e.g., Black or Japanese patients).
- **Photosensitivity** (moderate).
- **Scleroderma-like changes**: - Edematous sclerodermatous plaques affecting extremities, **sparing hands and feet**. - Other reported forms: plaque-type morphea, guttate morphea, atrophoderma of Pasini and Pierini, lichen sclerosus et atrophicus.

Other Signs:

- **Mousy or musty odor** in urine and sweat due to excretion of **phenylacetic acid**.
- **Vomiting** in infancy.

Diagnosis

- **Elevated serum phenylalanine levels**: typically ≥ 20 mg/dL (10–50 times normal).
- **Normal or elevated plasma tyrosine levels**.

- **Increased urinary phenylpyruvic acid**, detectable by ferric chloride test (turns green with phenols).
- Newborn screening programs routinely test for hyperphenylalaninemia using blood spot analysis (Guthrie test or tandem mass spectrometry).

Note: The image referenced (FIGURE 131-4) illustrates diffuse plantar hyperkeratosis in tyrosinemia and its resolution on a low-tyrosine, low-phenylalanine diet, highlighting the importance of metabolic dietary management in related disorders.